2.9 Dimension and Rank

Def: The dimension of a nonzero subspace H, denoted by $\dim H$, is the number of vectors in any basis for H. The dimension of the zero subspace $\{0\}$ is defined to be zero.

- The space $\mathbb{R}^n$ has dimension n. Every basis for $\mathbb{R}^n$ consists of n vectors.
- To find the dimension of $\text{Nul}\,A$, simply identify and count the number of free variables in $Ax = 0$.

Def: The rank of a matrix A, denoted by $\text{rank}\,A$, is the dimension of the column space of A.

- Since the pivot columns of A form a basis for $\text{Col}\,A$, the rank of A is just the number of pivot columns in A.

$$\dim \text{Col}\,A = \text{rank}\,A$$

Ex: Determine the rank of the matrix

$$A = \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 4 & 7 & -4 & -3 & 9 \\ 6 & 9 & -5 & 2 & 4 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix}$$

Sol: Reduce A to echelon form:

$$A \sim \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 4 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

pivot columns

The matrix A has 3 pivot columns, so $\text{rank}\,A = 3$.

Theorem: The Rank Theorem:

If a matrix A has n columns, then $\operatorname{rank} A + \dim \operatorname{Nul} A = n$.

i.e $\dim \operatorname{Col} A + \dim \operatorname{Nul} A = n$.

Theorem: The basis theorem:

Let H be a p-dimensional subspace of $\mathbb{R}^n$. Any linearly independent set of exactly p elements in H is automatically a basis for H. Also, for any set of p elements of H that spans H is automatically a basis for H.

Theorem: The Invertible Matrix Theorem (cont'd).

Let A be an $n \times n$ matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix.

m. The columns of A form a basis of $\mathbb{R}^n$.

n. $\operatorname{Col} A = \mathbb{R}^n$

o. $\dim \operatorname{Col} A = n$

p. $\operatorname{rank} A = n$

q. $\operatorname{Nul} A = \{0\}$

r. $\dim \operatorname{Nul} A = 0$

$A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}$.

If A is an $n \times n$ invertible matrix

Ex: Determine the dimension of the subspace H of $\mathbb{R}^3$ spanned by the vectors v_1, v_2 and v_3. (Plot, find a basis for H.)

$$v_1 = \begin{bmatrix} 2 \\ -8 \\ 6 \end{bmatrix} \quad v_2 = \begin{bmatrix} 3 \\ -7 \\ -1 \end{bmatrix} \quad v_3 = \begin{bmatrix} -1 \\ 6 \\ -7 \end{bmatrix}$$

Sol. A basis for this space is provided by the pivot columns of A.

$$A = \begin{bmatrix} 2 & 3 & -1 \\ -8 & -7 & 6 \\ 6 & 1 & -7 \end{bmatrix} \sim \begin{bmatrix} 2 & 3 & -1 \\ 0 & 5 & 2 \\ 0 & -10 & -4 \end{bmatrix} \sim \begin{bmatrix} 2 & 3 & -1 \\ 0 & 5 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

The first two columns of A are pivot columns and form a basis for H. Thus $\dim H = 2$.

Ex: Let $A = \begin{bmatrix} 1 & -1 & 5 \\ 2 & 0 & 7 \\ -3 & -5 & -3 \end{bmatrix}$ and $u = \begin{bmatrix} -7 \\ 3 \\ 2 \end{bmatrix}$

Is u in Nul A? ✓ Is u in Col A? ✗

Sol. To check whether u is Nul A, simply check $Au = 0$.

$$Au = \begin{bmatrix} 1 & -1 & 5 \\ 2 & 0 & 7 \\ -3 & -5 & -3 \end{bmatrix} \begin{bmatrix} -7 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

So, u is Nul A.

Deciding whether u is in Col A requires more work. You have to check whether $Ax = u$ has a solution. Reduce the augmented matrix $[A\ u]$ to echelon form to see whether the equation $Ax = u$ is consistent.

$$\begin{bmatrix} 1 & -1 & 5 & -7 \\ 2 & 0 & 7 & 3 \\ -3 & -5 & -3 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 5 & 7 \\ 0 & 2 & -3 & 17 \\ 0 & -8 & 12 & -19 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 5 & 7 \\ 0 & 2 & -3 & 17 \\ 0 & 0 & 0 & 49 \end{bmatrix}$$

not consistent

so u is not in Col A.